

Mohini Deshpande

AI Engineer building agentic systems, RAG pipelines, and production ML — from architecture through daily use.

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About

AI and Software Engineer specialising in applied GenAI and agentic AI systems, with production experience across backend engineering, LLM integration, and NLP. I build Python services, agentic workflows, and retrieval-augmented generation systems end to end, from requirements gathering through daily production use. My background spans model evaluation and fine-tuning, cloud deployment, and automation tooling, with a track record of full ownership on solo and freelance engagements.

Expertise

Languages & Frontend

Python, JavaScript/TypeScript, SQL, MySQL, React, HTML, CSS

Backend & Data

FastAPI, Node.js, REST APIs, PostgreSQL, Redis, Celery, MongoDB, vector databases, Qdrant, pgvector

AI, LLM & Agentic Systems

Google ADK, Gemini API, Llama API, LangChain, Hugging Face, Model Context Protocol, retrieval-augmented generation, prompt engineering, multi-agent orchestration

Machine Learning & NLP

PyTorch, TensorFlow, neural networks, Transformer architecture, fine-tuning, BERT, FinBERT, tokenization, Named Entity Recognition

Cloud, DevOps & Tools

AWS, GCP, Azure ML, Docker, Kubernetes, CI/CD, GitHub Actions, Git, Claude Code, GitHub Copilot

Automation & Data

n8n, Make, Zapier, UiPath, Microsoft Copilot, Excel data cleaning and reconciliation

Experience

Contract Software Engineer — *Ekblink* Jul 2026 – Present

- Building backend infrastructure using Python, FastAPI, Redis, and PostgreSQL for a production AI content-moderation pipeline, with logging and monitoring for observability.

Contract Software Engineer — *Independent Client, India* Dec 2024 – Mar 2025

- Took full ownership as sole engineer of a Python and Gemini API agent that processes unstructured inbound client emails, auto-drafting responses to roughly 65–70% of the 45–50 weekly inquiries and routing edge cases to a human.
- Introduced pytest and DeepEval tests to validate model output against expected schemas; the agent has run in daily production use since I built and shipped it.

AI Trainer — *Outlier, Remote* Oct 2024 – Jul 2025

- Designed evaluation frameworks to assess and reduce bias across 150+ AI model responses, lifting output credibility by 20%, and translated findings into clear recommendations for technical and non-technical stakeholders.

Machine Learning Intern — *SmartKrawler, India* Apr – May 2021

- Built Python ETL pipelines and CNN/LSTM neural network models, lifting data quality by 25% and improving business metric projections by 12%.

Projects

FinBERT & LSTM Market Intelligence — *NLP sentiment & forecasting pipeline*

- Built an end-to-end NLP pipeline fine-tuning a pretrained FinBERT transformer model for sentiment classification on financial news, reaching 88% accuracy, then feeding those signals into an LSTM/MLP forecasting stage at 75% precision.

FixIt AI — *Autonomous agentic assistant* github.com/MohiniDeshpande/ai-handyman-backend

- Architected an agentic assistant in Python using Google's Agent Development Kit and a FastAPI backend, combining LLM prompting and function/tool calling with a RAG pipeline over Qdrant and pgvector for grounded, context-aware responses.

PokéMarket — *AI market intelligence platform* github.com/MohiniDeshpande/pokemarket-ai

- Built a React frontend for a full-stack agentic chatbot, integrating MongoDB MCP and the Gemini API to deliver real-time profit tracking and financial metrics, containerised with Docker and deployed via CI/CD.

Education

MSc Advanced Computer Science: AI

University of Manchester, UK — 2023–2024

Dissertation: Animating Avatars Using Motion Capture Data — motion graphs, clustering, point cloud processing

Coursework: Reinforcement Learning, Deep Learning, NLP, Computer Vision, Data Engineering

BSc Computer Science

University of Mumbai, India — 2019–2022

Dissertation: Restaurant Recommendation System Using Machine Learning and Flask

Certifications

Google Cloud Platform Fundamentals — Coursera

ML Pipelines with Azure ML Studio — Coursera

UiPath Workflow Automation Certification

Ethics of AI — University of Helsinki, in progress